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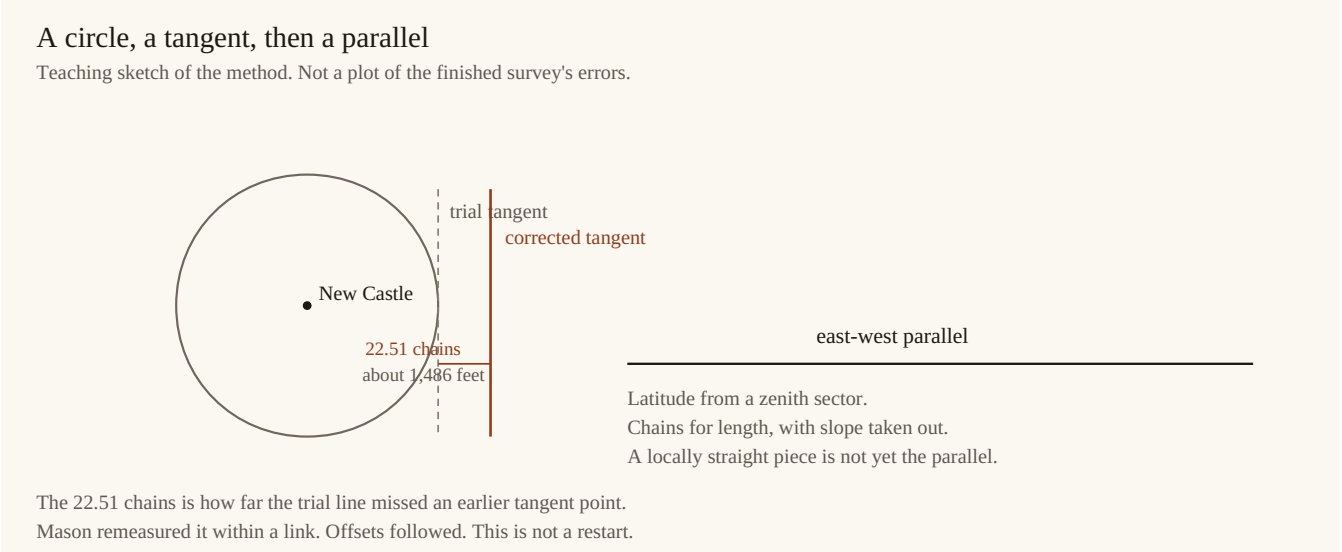
The Mason-Dixon survey: how a boundary became real

The Mason-Dixon line is not one endless straight stroke. The Penn and Calvert grant needed a circle around New Castle, a tangent across the Delmarva peninsula, and then an east-west parallel. Each piece answers a clause in a charter. None of them is "draw west until the paper runs out."

Latitude came from a zenith sector, a telescope pointed near the zenith so the star's height gives the parallel. Length came from chains of a known size, with slope leveled out. On a round Earth a short piece that looks straight on the ground is not yet a parallel of latitude. You compute the parallel, then you step the offsets that keep you on it.



The New Castle circle, a trial tangent, and the parallel

A sketch of the method, drawn for this lesson. It is not a graph of leftover error along the finished line.

In August 1764 a trial tangent about 81 miles long missed an earlier colonial tangent point by 22.51 chains, about 1,486 feet. Read that carefully. It is the offset of a planned trial from an older mark, not the collapse of a finished survey. Mason took a man and remeasured the gap himself. He came within a link, about eight inches. Through the autumn they laid proportional offsets and declared a true tangent. They did not burn the work and begin again. No document found shows one single gross mistake that forced a full restart. That sentence does not prove the party never erred. Later resurveys, in the nineteenth and twentieth centuries, are different instruments and different legal questions.

On 17 September 1767 a falling tree killed the pack-horse drivers William Baker and John Carpenter. The names come from a careful later account of the survey; they are not written in Mason's own journal, so the lesson keeps that distinction. On 9 October the leader of the escort, who had joined in July with Mohawk and Onondaga men, said the war path was as far as the chiefs

of the Six Nations had commissioned him to go. He would not proceed one step farther west. Earlier, twenty-six hired men had refused a river for fear of the Shawnee and the Delaware. The journal states the commission's limit. It does not, in its own words, explain why the chiefs drew it there. Later markers that say hostile nations "ordered them to halt" are an interpretation, not the 1767 sentence.

The line later became a symbol of North and South. That symbol is a nineteenth-century use of an eighteenth-century colonial boundary. Dixon's lesson has the man who helped lay it. The geometry here is only the geometry.

The illustrated lesson is download the PDF.

Sources

- Todd Babcock, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_babcock.pdf
- Edwin Danson, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_danson.pdf
- Mason and Dixon, 1767 letter: https://peterschmidt.domains.swarthmore.edu/engl52b/m-d.letter.html

Key facts

- The boundary is a New Castle circle, a Delmarva tangent, and an east-west parallel, not one long line.
- The 1764 trial tangent missed an older point by 22.51 chains. Mason remeasured within a link, and offsets corrected the line. It was not a documented full restart.
- On 9 October 1767 the escort stopped at the war path, the limit of the Six Nations commission.
- Baker and Carpenter were killed by a falling tree on 17 September 1767 in later accounts of the work, not in Mason's journal itself.